2. Enable query execution plan logging for slow queries using auto_explain
3. Find matching queryid values in the postgresql.log

Make sure compute_query_id is set to on in your PostgreSQL config file.

Once you've found a match, you have the query as executed with real parameters that you can analyze using PostgreSQL EXPLAIN.

You now have a game plan to find high frequency and slow queries with PGSS, and then either manually or automatically collect samples from the log, including manual or automatic collection of the query execution plan. You haven't yet seen how to use the query plan, but that will be coming soon.

You're just about ready to dive into execution plans.

Before doing that, there's one more thing to consider before working on query optimization. PostgreSQL can require some database maintenance for tables and indexes that are related to the slow query. Before spending a lot of energy on query optimization, make sure the tables and indexes are optimized.

Performing Maintenance First

Before diving too far into query execution plans make sure the tables and indexes related to the query are optimized.

Perform some of the generic database maintenance steps when you're facing a specific slow query.

1. For each table in the query run a manual VACUUM (ANALYZE) from psql for the table. This makes sure Vacuum has run recently and statistics are updated.
2. For any indexes used by the query, determine whether they have a high percentage of estimated bloat. If so, rebuild the index.
3. If you see Autovacuum or Analyze not running regularly for Indexes with very high bloat across the board, it may be time to schedule supplemental Vacuum, Analyze, and Reindex operations. You'll learn more about that later.

Refer to the *Show database bloat*[16] scripts from the PostgreSQL Wiki. These scripts help you identify indexes with a very high estimated percentage of bloat.

Now that you've made sure the tables and indexes are in good shape, you're ready to dive into query optimization.

16. https://wiki.postgresql.org/wiki/Show_database_bloat